

SMART GRAIN STORAGE CONDITION MONITORING AND SPOILAGE RISK PREDICTION FOR PDS USING AI AND IOT

GOKUL KANNAN - 24619113

VIGNESH - 24619157

UMESH - 24619161

VELS INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)

MINI PROJECT REPORT

Submitted to the

SCHOOL OF ENGINEERING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

in Partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

VELS INSTITUTE OF SCIENCE, TECHNOLOGY AND ADVANCED STUDIES (VISTAS)

CHENNAI – 600 117

BONAFIDE CERTIFICATE

Certified that this project report "SMART GRAIN STORAGE CONDITION MONITORING AND SPOILAGE RISK PREDICTION FOR PDS USING AI AND IOT", is the Bonafide work of GOKUL KANNAN - 24619113, VIGNESH - 24619157, UMESH – 24619161 who carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE OF THE HOD
Dr. K. Ulaga Priya M.E., Ph.D
Associate Professor & Head
Of Department AI&ML
School of Engineering
VISTAS
Chennai

SIGNATURE OF THE SUPERVISOR
Ms.S.J.Kavitha, M.E, (Ph.D)
Assistant Professor
Department of AI&ML
School of Engineering
VISTAS
Chennai

Submitted for the Project Work and Viva - Voce examination held on …… at Vels Institute of Science, Technology and Advanced Studies (VISTAS), Chennai – 600 117.

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

It is certainly a great delight and proud privilege to acknowledge the help and support I received from the positive minds around me in making this venture a successful one. The infrastructural support with all kinds of lab facilities have been a motivating factor in this completion of project work, all because of FOUNDER & CHAIRMAN, Shri. Dr. ISHARI K. GANESH., with great pleasure that I take this opportunity to thank him. I am grateful to our HEAD OF THE DEPARTMENT, Dr. K. Ulaga Priya, for her continuous support for the project, from initial advice and contacts in the early stages of conceptual inception, and through on-going advice and encouragement to this day. I extend our warmest thanks to my INTERNAL GUIDE, Ms. S. J. Kavitha, Ass.Prof/AI&ML, who has patiently guided and provided us with invaluable advice and help. I would like to express my greatest gratitude to our Staff members, Librarian, and Non-teaching staff members of Computer Centre, who have helped and supported me throughout. Last but not the Least, I wish to thank my parents and my friends for their individual support and interest who has inspired me and encouraged me to go in my own way, without them I would be unable to complete this project.

GOKUL KANNAN - 24619113

VIGNESH - 24619157

UMESH - 24619161

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ABSTRACT

Food grain spoilage during storage is a major challenge in the Public Distribution System (PDS) due to environmental factors such as high temperature, humidity, excess moisture, and poor air quality, which reduce grain quality and lead to food wastage and economic loss. This project proposes a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System using Artificial Intelligence (AI) and the Internet of Things (IoT) to continuously monitor storage conditions and provide early warnings before spoilage occurs. The proposed system uses an ESP32 microcontroller integrated with a DHT22 temperature and humidity sensor, grain moisture sensor, and MQ135 gas sensor to collect real-time environmental data from grain storage facilities. The collected data is transmitted to an IoT cloud platform for continuous monitoring and analysis. A Machine Learning model processes the sensor data and predicts the spoilage risk by classifying the storage condition as Safe, Moderate Risk, or High Risk, enabling warehouse authorities to take timely preventive actions. The proposed solution aims to reduce food wastage, improve grain storage efficiency, minimize manual inspection, and support smart warehouse management in PDS godowns through a low-cost, scalable, and intelligent monitoring system.

Keywords: Artificial Intelligence, Internet of Things, Smart Grain Storage, Spoilage Risk Prediction, ESP32, Grain Moisture Monitoring, Public Distribution System

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ANN	Artificial Neural Network
CCT	Compact Convolutional Transformer
CSS	Cascading Style Sheets
CSV	Comma-Separated Values
GSM	Global System for Mobile Communications
I2C	Inter-Integrated Circuit
IDE	Integrated Development Environment
IoT	Internet of Things
LCD	Liquid Crystal Display
LED	Light Emitting Diode
ML	Machine Learning
PDS	Public Distribution System
RF	Radio Frequency
SMS	Short Message Service
WSN	Wireless Sensor Network

CHAPTER 1
INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Food grains are essential commodities in the Public Distribution System (PDS), which helps make staple food available to the public. Before these grains are distributed, they are stored in warehouses or godowns for a period of time. Maintaining suitable storage conditions is important because deterioration during storage can affect grain quality, increase wastage, and cause economic loss. A storage facility therefore needs reliable information about the conditions surrounding the stored grains, rather than depending only on occasional observation.

The project, “Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction for PDS using AI and IoT,” proposes a low-cost system for monitoring important storage conditions and estimating the risk of grain spoilage. It combines sensors, an ESP32 microcontroller, wireless connectivity, an IoT dashboard, and a machine-learning model. The intended purpose is to help warehouse staff identify potentially unsafe conditions earlier and take preventive action before the quality of stored grain is seriously affected.

The project focuses on environmental indicators associated with grain storage: temperature, relative humidity, grain moisture, and air quality. High temperature, excessive humidity, excess moisture, and poor air quality can be warning signs that storage conditions need attention. By collecting these readings continuously, the proposed system aims to provide a more consistent view of the storage environment than manual checks alone.

This chapter introduces the storage problem, the proposed monitoring approach, the role of its main components, and the scope and objectives of the project. The description follows the project abstract and project presentation; it presents the proposed design and intended functions, not a claim that large-scale warehouse deployment or predictive accuracy has already been validated.

1.1.1 Background and Context

Public Distribution System warehouses must hold food grains safely until they are needed for distribution. During storage, conditions may change with weather, ventilation, the moisture content of the grain, and the characteristics of the storage location. If such changes are not noticed in time, grain may develop problems such as fungal growth, insect infestation, bad odour, or other forms of deterioration. These outcomes can reduce the usable quality of grain and contribute to food wastage.

Traditional monitoring often depends on staff inspecting the warehouse and checking conditions at intervals. Manual inspection remains useful, but it cannot continuously represent changes between inspections. A condition that becomes unsuitable after a routine check may

remain unnoticed until the next visit. Recording observations manually can also make it harder to compare readings over time or identify a developing pattern.

A monitoring system based on sensors can help address this gap by taking repeated measurements and making them available in one place. IoT connectivity can carry readings from the warehouse to a dashboard so that an authorised user can review conditions without needing to stand beside each sensor. When the measurements are considered together, they can offer a more useful indication of storage status than any single reading on its own.

The project applies this approach to PDS grain storage. Its proposed system is designed to observe selected environmental indicators and present a risk category—Safe, Moderate Risk, or High Risk. These categories are intended to make readings easier to interpret and support timely decisions by warehouse staff. They are warning categories for the proposed system and should not be treated as a substitute for physical inspection or established grain-quality procedures.

1.1.2 Problem Statement

Food grain stored for long periods can be affected by changes in temperature, humidity, moisture, and air quality. Excessive values or unfavourable combinations of these conditions may contribute to deterioration, including fungal growth, insect activity, and bad odour. The consequences may include lower grain quality, food wastage, and financial loss. For PDS warehouses, early awareness of changing storage conditions is therefore important.

The key problem addressed by this project is the limited ability of periodic manual inspection to provide continuous, timely information about the storage environment. If environmental changes occur between inspections, staff may not receive an early indication that conditions are becoming risky. In addition, readings collected at different times may not be available through a single real-time view for convenient monitoring.

The proposed solution is an integrated IoT and AI-based monitoring arrangement. Sensors measure temperature, humidity, grain moisture, and air quality. An ESP32 acts as the main controller and gathers the sensor readings. The readings are then transmitted over Wi-Fi to an IoT dashboard for viewing. A machine-learning component is intended to analyse the collected measurements and classify the storage condition into Safe, Moderate Risk, or High Risk.

The system is intended to provide an early warning and decision-support capability. It does not physically remove moisture, control ventilation, or prevent spoilage by itself. Its role is to help identify conditions that may require attention so that responsible personnel can inspect the warehouse and take suitable preventive measures.

1.1.3 Need for the Proposed System

Continuous visibility is useful because warehouse conditions can change over time. A digital record of temperature, humidity, grain moisture, and air-quality readings can help staff notice

changes that may be missed during occasional inspection. A dashboard can also make the latest readings easier to access and interpret. These features are particularly relevant when a warehouse needs regular monitoring but resources for frequent manual checks are limited.

The proposed design uses commonly available, low-cost electronic components. The ESP32 provides a central controller and Wi-Fi connectivity. The DHT22 measures temperature and relative humidity; a moisture sensor measures grain moisture; and the MQ135 sensor is used to monitor air quality through gas-related readings. The values are collected for display and analysis. A 16×2 I2C LCD, green, yellow and red LEDs, and a buzzer serve as local display and alert components.

The AI component adds a risk-estimation step to the monitoring process. Instead of presenting sensor values alone, the model is intended to map the collected data to an understandable storage-risk category. This can help a user recognise when the readings indicate a condition that deserves closer attention. The quality of such predictions depends on the data and model used, so suitable training data and testing are important parts of developing the system.

Together, the monitoring and prediction functions aim to reduce dependence on manual observation alone, improve awareness of storage conditions, and support preventive action. The system is proposed as a practical prototype that can be extended or evaluated further before wider use in actual PDS warehouses.

1.1.4 Overview of the System Operation

The proposed workflow begins when the sensors collect measurements from the grain-storage environment. The ESP32 reads the connected sensors and acts as the central controller. The collected readings are made available through Wi-Fi to an IoT dashboard, where a user can view the current storage conditions. The same measurements can be prepared for analysis by the machine-learning component, which predicts a risk category for the monitored conditions.

The DHT22 sensor provides temperature and relative-humidity readings. The grain-moisture sensor supplies an indication of moisture in the stored grain. The MQ135 sensor provides an air-quality-related measurement. Each sensor contributes a different piece of information; considering these readings together is intended to provide a broader picture of the environment than monitoring only temperature or humidity.

The IoT dashboard serves as the remote monitoring interface. It displays the incoming readings in a way that users can review. Local indicators such as an LCD, LEDs, and a buzzer provide nearby status information and alerts. The AI/ML stage classifies the condition as Safe, Moderate Risk, or High Risk, enabling a user to decide whether further inspection or preventive action is needed.

At a high level, the sequence is: sensors collect grain-storage data; the ESP32 reads and processes the sensor values; Wi-Fi sends the values to the dashboard; the AI model estimates

the spoilage-risk category; and the system communicates the status or alert. This flow forms the basis for the proposed architecture described in the later chapters of the report.

1.2 SCOPE AND OBJECTIVE

1.2.1 Aim of the Project

The main aim of this project is to design a smart grain-storage condition monitoring and spoilage-risk prediction system for PDS warehouses by combining Internet of Things (IoT) sensing with Artificial Intelligence (AI) and machine learning. The system is intended to collect key environmental readings continuously, make those readings accessible through a dashboard, and estimate the storage condition using clear risk categories.

The project concentrates on monitoring and early warning. It is not designed as an automated grain-treatment or warehouse climate-control system. Its purpose is to provide information that can help warehouse personnel recognise potentially unfavourable conditions and take suitable action. The effectiveness of this support depends on correct sensor installation, reliable readings, appropriate model training, and meaningful evaluation.

1.2.2 Primary Objective

The primary objective is to develop an AI- and IoT-based prototype that monitors important grain-storage conditions and predicts spoilage risk for PDS storage environments. It brings together data collection, wireless transmission, dashboard presentation, and risk classification in a single proposed workflow.

1.2.3 Specific Objectives

- To monitor the temperature inside the grain-storage environment.
- To measure relative humidity continuously using a suitable sensor.
- To measure grain moisture as an indicator relevant to storage condition.
- To monitor air-quality-related changes using the MQ135 sensor.
- To use the ESP32 as the central controller for reading sensor data.
- To transmit readings over Wi-Fi to an IoT dashboard for remote viewing.
- To apply a machine-learning model to classify conditions as Safe, Moderate Risk, or High Risk.
- To provide readable status indications and alerts that can support preventive action.
- To contribute to reducing grain spoilage and food wastage through earlier awareness of risky conditions.

1.2.4 Functional Scope

The functional scope covers four connected activities: sensing, communication, analysis, and notification. In the sensing stage, the system collects temperature, relative humidity, grain-

moisture, and air-quality-related readings. The sensors are connected to the ESP32, which acts as the main controller and coordinates collection of the data.

In the communication stage, the ESP32 uses Wi-Fi to send readings to an IoT platform. The prototype presents the readings on a web-based dashboard (shown in the project results as an ESP32 local monitoring and control dashboard). The dashboard makes the current readings available in a convenient form so that a user can monitor the storage environment, subject to network availability and the implemented configuration.

In the analysis stage, a machine-learning model processes the collected measurements and assigns a storage-risk category. The project uses Python and Google Colab for AI/ML development and analysis using a sensor dataset. The risk labels are Safe, Moderate Risk, and High Risk. The model must be trained and tested using suitable data before its predictions can be relied upon; this report does not assume a specific accuracy level that has not been established by the supplied project materials.

In the notification stage, the storage condition is communicated through the dashboard and, where implemented, local components such as an LCD, LEDs, and a buzzer. The alerts are intended to draw attention to readings or classifications that may need review. Final decisions about grain handling remain with the responsible personnel.

1.2.5 Hardware Scope

The proposed hardware includes an ESP32 microcontroller, a DHT22 temperature-and-humidity sensor, a grain-moisture sensor, an MQ135 air-quality sensor, and local output components. The ESP32 is the central unit that reads sensor inputs and supports wireless communication. It is used as the controller of the prototype.

The DHT22 is used to obtain temperature and relative-humidity values from the storage environment. These measurements are relevant because unsuitable temperature and humidity can be associated with deterioration risk. The grain-moisture sensor is intended to provide information about the moisture level of the grain, while the MQ135 sensor provides an air-quality-related signal that can be monitored as another indicator of changing storage conditions.

An LCD may be used to show local readings or status, while LEDs and a buzzer can provide visual and audible indications. The exact presentation depends on the final wiring and program implementation. The hardware scope is limited to monitoring and alerting functions described in the project plan; automatic drying, ventilation control, pest treatment, and other physical interventions are outside the stated scope.

1.2.6 Software and Data Scope

Arduino IDE is used for programming the ESP32 and integrating the sensors, display, indicators, and Wi-Fi communication. A web-based dashboard is used for viewing readings. Python and Google Colab are used for preparing and analysing a dataset and developing the

machine-learning component. These tools support the flow from physical measurements to remote monitoring and risk estimation.

The model's predictions depend on the data used for training and testing. The supplied project material describes the approach but does not establish a fully validated, long-term dataset from PDS warehouses. Therefore, the project should treat the prediction stage as a prototype capability and evaluate it with representative data before using it for operational decisions.

1.2.7 Expected Outcomes and Benefits

The expected outcome is a working prototype that brings together sensor-based monitoring, wireless transmission, dashboard display, and machine-learning-based risk classification. It is intended to make selected storage conditions visible in real time and to present a simplified risk category that can be understood by warehouse personnel. Local display and alert components may provide additional notification near the monitoring unit.

The expected benefits include improved awareness of changing storage conditions, reduced reliance on occasional manual checks as the only source of information, and earlier identification of conditions that may require inspection. By helping personnel respond sooner, the system aims to support better grain-storage management and contribute to reducing avoidable food wastage. These are intended benefits and should be assessed through prototype testing and practical evaluation.

The project also provides an integrated learning platform for embedded systems, sensors, IoT connectivity, dashboard development, and machine learning. It demonstrates how measurements from physical sensors can be sent to a digital interface and then used by an analytical model to produce a simple decision-support output.

1.2.8 Limitations and Boundaries

The system monitors selected indicators rather than every factor that can affect grain quality. Sensor readings may be influenced by placement, calibration, maintenance, and the characteristics of the storage environment. Air-quality-related readings should be interpreted according to the sensor's capabilities and should not be treated as a direct laboratory confirmation of spoilage.

The risk category is only as dependable as the training data, input quality, and model evaluation. Conditions in different warehouses, grain types, seasons, and locations may vary. Before operational deployment, the prototype would need appropriate calibration, testing with representative measurements, and validation against reliable assessments of grain condition. Network or power interruptions may also affect continuous monitoring and dashboard updates.

The project does not claim to replace warehouse staff, laboratory testing, or established storage procedures. It is intended to supplement them with continuous measurements and an early-warning aid.

1.2.9 Summary of Scope

In summary, the project scope includes monitoring temperature, humidity, grain moisture, and air-quality-related readings; collecting them through an ESP32-based sensor unit; transmitting them through Wi-Fi to an IoT dashboard; and using a machine-learning model to classify the monitored condition as Safe, Moderate Risk, or High Risk. Local indicators such as an LCD, LEDs, and a buzzer are part of the alert arrangement.

The scope is centred on a low-cost, connected prototype for PDS grain-storage environments. It includes hardware integration, embedded programming, dashboard-based monitoring, and AI/ML analysis. It does not include automatic correction of environmental conditions or a claim of proven deployment at warehouse scale. Further validation would be needed to establish the model's reliability across real storage settings.

1.2.10 Chapter Conclusion

This chapter has introduced the need for improved grain-storage monitoring in the Public Distribution System and described the proposed Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System using AI and IoT. The problem arises from the possibility that temperature, humidity, grain moisture, and air-quality conditions may become unfavourable between manual inspections. The proposed system addresses this need by combining sensor measurements, ESP32-based collection, Wi-Fi communication, an IoT dashboard, and machine-learning risk classification.

The objectives and scope establish the system's intended role as a monitoring and early-warning tool. By making readings accessible and presenting a clear risk category, the prototype aims to support timely inspection and preventive action. The following chapters present the literature review, system design, implementation, and conclusion of the project.

CHAPTER 2
LITERATURE REVIEW

2.1 LITERATURE OVERVIEW

Monitoring the condition of stored grain has been studied through sensor-based measurement, Internet of Things (IoT) and wireless sensor network (WSN) architectures, and machine-learning (ML) and artificial-intelligence (AI) methods for predicting grain quality. The variables most often monitored are grain moisture, temperature, relative humidity and gases produced during storage, such as carbon dioxide (CO_2), because changes in these quantities can indicate deterioration before it becomes visible.

This chapter reviews ten studies published between 2022 and 2024 that are relevant to the proposed Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System. They include review papers, IoT prototypes, wireless sensor network platforms, and ML

models for predicting quality, moisture or contamination of stored grain. The review serves the following purposes:

- To understand how sensors, IoT platforms and AI/ML techniques have been combined for stored-grain monitoring.
- To compare the methodologies used and the limitations reported or evident in each study.
- To identify the research gap with respect to low-cost, AI-enabled monitoring for Public Distribution System (PDS) warehouses.
- To provide the technical basis for the design of the proposed ESP32-based prototype.

2.2 LITERATURE SURVEY

The surveyed work is grouped below by approach. Reference numbers correspond to the list of references at the end of this report.

Reviews of Sensors, IoT and Artificial Intelligence for Stored Grain

Lutz and Coradi [1] reviewed sensor-based monitoring, IoT and WSN architectures, and machine-learning approaches for stored-grain quality, covering variables such as moisture, temperature, relative humidity and intergranular CO_2. Their review explains how real-time sensing and ML can support early detection of deterioration and better storage decisions, and therefore provides a technical basis for combining sensors, IoT and AI in grain storage. As a review, however, it is not a field-validated PDS system, and no single integrated spoilage-risk model designed specifically for Indian PDS warehouses is presented.

Priyadarsani, Pradhan and Jena [6] reviewed IoT applications for food-grain storage in the Indian context, including real-time monitoring, early warning, environmental control, inventory, traceability and predictive analytics. The study connects IoT-based storage technology directly with Indian food-security concerns such as spoilage, mold and insect infestation. It does not experimentally validate a complete AI and IoT PDS warehouse system or quantify predictive performance on PDS datasets.

IoT-Based Warehouse and Storage Monitoring Prototypes

Lydia et al. [2] presented an automated food-grain monitoring system for warehouses using IoT. The system uses sensor nodes with a PIC microcontroller and GSM/cloud connectivity, and reports through mobile and SMS messages. It monitors grain level, temperature, humidity, vibration, CO, motion and smoke. This supports remote warehouse monitoring and reduces dependence on manual inspection, but the monitoring is mainly threshold-based, predictive AI or spoilage-risk estimation is limited, and large-scale, long-duration PDS validation is absent.

Pratama, Nurchim and Pamekas [9] developed a prototype of smart grain storage based on IoT using a design-based research approach. The prototype uses an ESP32, DHT22 temperature

and humidity sensors, an LDR and soil-moisture sensing, with cloud data transmission and early warnings. It provides a low-cost IoT architecture for real-time grain-storage monitoring, which is close to the hardware approach of the present project. The authors note that the prototype mainly monitors conditions, that automatic control and mobile integration were future improvements, and that AI-based spoilage-risk prediction is not its main focus.

Wireless Sensor Networks and Neural-Network Prediction for Corn

Coradi et al. [3] developed a prototype wireless sensor network and IoT platform for real-time monitoring of intergranular equilibrium moisture content, with neural-network models predicting the physical, physicochemical and microbiological quality of corn stored in silo bags. Lutz et al. [4] similarly used a WSN and IoT platform for real-time temperature and relative-humidity monitoring, with artificial neural networks (ANN) predicting grain quality under different moisture, temperature and packaging conditions. Together these studies show that equilibrium moisture content can be an important indicator of quality deterioration and that continuous monitoring can replace occasional sampling. Both are focused on corn, with limited storage setups or controlled experimental conditions, so validation across PDS grains, warehouse layouts and Indian climatic conditions is still needed.

Intergranular Monitoring of Wheat

Leal et al. [5] monitored temperature, relative humidity and CO_2 using sensors, and applied artificial neural networks and multiple linear regression to predict dry-matter loss in stored wheat. The work shows that changes in intergranular temperature, humidity and CO_2 can provide early information about grain-quality changes and support predictive decisions. It focuses on vertical-silo wheat storage and dry-matter loss rather than complete spoilage classification, and pest, fungal and operational variables are not fully integrated.

Machine Learning for Contamination and Moisture Estimation

Kim et al. [7] used a Compact Convolutional Transformer (CCT) to classify wheat as healthy, incipient or contaminated using CO_2 respiration and visual mold information, with a reported accuracy of 83.33%. The study demonstrates the value of distinguishing early-stage spoilage from already contaminated grain, but it uses a relatively small laboratory image dataset under controlled conditions, and more diverse field data and sensor/IoT integration are required for PDS deployment.

Azmi et al. [8] used 2.4 GHz and 868 MHz wireless (RF) measurements with an ensemble ML approach to estimate rice moisture content and localize wet spots, with a reported ensemble accuracy of about 94.8%. This supports non-invasive moisture estimation and localization of potential wet or spoilage areas in stored rice. The validation is testbed-based, and large heterogeneous PDS warehouses, different grain types and changing RF environments need further validation.

Machine-Learning Models for Corn Quality Across Post-Harvest Stages

Rodrigues et al. [10] monitored temperature, relative humidity and CO_2 during transport, drying and storage of corn, and compared regression, artificial neural network, decision tree and random forest models for predicting grain quality. The study integrates sensor measurements across post-harvest stages and compares ML models for predicting quality changes. It is focused on corn and a specific workflow; a PDS implementation would need continuous warehouse data, multi-grain validation, alerts, traceability and operational integration.

Table 1 – Summary of the reviewed literature

S.No.	Reference	Methodology	Limitation
1	Lutz and Coradi (2022) [1]	Review of sensors, IoT/WSN and ML for stored-grain quality	Review study; not a field-validated PDS system
2	Lydia et al. (2022) [2]	IoT warehouse monitoring with PIC microcontroller, GSM/cloud, SMS reporting	Mainly threshold-based; limited predictive AI
3	Coradi et al. (2022) [3]	WSN and IoT platform with neural networks for corn in silo bags	Corn only; limited storage setup
4	Lutz et al. (2022) [4]	WSN and IoT with ANN for grain quality under varied conditions	Controlled conditions; mainly corn
5	Leal et al. (2023) [5]	Sensors with ANN and multiple linear regression for wheat dry-matter loss	Vertical-silo wheat; no full spoilage classification
6	Priyadarsani et al. (2023) [6]	Review of IoT for food-grain storage in India	Review; no experimental validation
7	Kim et al. (2024) [7]	CCT classifier using CO_2 respiration and mold information for wheat	Small laboratory dataset
8	Azmi et al. (2024) [8]	RF measurements with ensemble ML for rice moisture	Testbed-based validation
9	Pratama et al. (2024) [9]	ESP32, DHT22, LDR and soil-moisture IoT prototype with early warning	Monitoring focus; AI prediction not the main focus
10	Rodrigues et al. (2024) [10]	Regression, ANN, decision tree and random forest for corn quality	Corn and a specific workflow

Research Gaps Identified

From the review, the following gaps emerge:

- Several of the studies are reviews or are focused on a single grain such as corn, wheat or rice, in silos, laboratory setups or testbeds, rather than on PDS warehouses.

- Low-cost IoT prototypes are mainly threshold-based monitors, and predictive AI or spoilage-risk estimation is limited.
- Studies that apply AI do not provide a low-cost, integrated solution for Indian PDS warehouses combining temperature, humidity, grain moisture and air quality with real-time alerts.
- Large-scale, long-duration validation across grain types, warehouse layouts and climatic conditions is absent.

The existing journal papers mainly focus on maize grain monitoring, anomaly detection, or advanced research technologies such as electronic-nose systems. The proposed project addresses this gap by developing an ESP32-based IoT prototype that monitors temperature, humidity, grain moisture and air quality, while an AI model predicts the storage condition as Safe, Moderate Risk or High Risk and real-time alerts are provided through an IoT dashboard and local indicators. This makes the proposed system practical, affordable and suitable for PDS warehouse management.

Summary of Literature Findings

The reviewed literature shows that sensor-based monitoring, IoT connectivity and ML prediction can each support early detection of grain deterioration, and that temperature, relative humidity, moisture and gas concentration are meaningful indicators. At the same time, most of the studies concentrate on particular grains or laboratory conditions, and few combine low-cost hardware, AI-based risk classification and real-time alerts for PDS storage. These findings form the motivation for the proposed system described in the next chapter.

CHAPTER 3
SYSTEM DESIGN

3.1 EXISTING SYSTEM

Overview

In existing grain storage practice, the condition of the grain is mainly checked by manual inspection and traditional monitoring methods. Farmers and warehouse workers examine the grain periodically, which is time-consuming and may not detect spoilage at an early stage. Public Distribution System warehouses store essential food grains such as rice, wheat and dhal for long periods before distribution, so a condition that develops between two inspections can remain unnoticed.

Limitations of the Existing System

- Grain quality is monitored manually by workers.
- Temperature and humidity are not monitored continuously.

- Spoilage gases cannot be detected in real time.
- High moisture levels may lead to fungal growth and grain damage.
- There is no automatic alert or prediction system.
- Traditional methods increase the risk of grain spoilage and food loss.

High temperature, excessive humidity, increased grain moisture and poor air quality can lead to fungal growth, insect infestation, bad odor and grain spoilage. Because the existing system depends on periodic observation, it cannot provide continuous information or early warning, which can result in food wastage, reduced grain quality and economic loss. There is therefore a need for a low-cost system that monitors storage conditions continuously and warns staff before spoilage occurs.

3.2 PROPOSED SYSTEM

Overview

The proposed system is a low-cost, real-time IoT and AI system for monitoring the environmental conditions inside a PDS grain warehouse. An ESP32 microcontroller collects temperature and humidity from a DHT22 sensor, grain moisture from a moisture sensor, and air quality from an MQ135 gas sensor. The data are made available on a web dashboard and are analyzed by a Python-based AI/ML model, which classifies the storage condition as Safe, Moderate Risk or High Risk. Alerts are produced through an LCD display, LED indicators, a buzzer and the dashboard so that warehouse staff can take preventive action before spoilage occurs.

Features of the Proposed System

- Continuous monitoring of temperature, relative humidity, grain moisture and air quality.
- ESP32-based collection of sensor data with initial processing and filtering.
- Wi-Fi transmission of readings for real-time viewing on a dashboard.
- AI-based classification of storage condition into Safe, Moderate Risk and High Risk.
- Condition analysis and recommendations such as Dry Grains, Improve Ventilation and Inspect Storage.
- Local alerts through a 16×2 LCD, green/yellow/red LEDs and a buzzer.
- A dashboard that shows sensor values, system status, and automatic or manual operating modes.

Advantages

- Continuous monitoring replaces occasional manual checks as the only source of information.

- Early warning of unfavorable conditions supports preventive action and helps reduce grain spoilage and food wastage.
- Low-cost, commonly available components make the prototype affordable and scalable.
- Risk categories and recommendations make the readings easier for warehouse staff to interpret.
- Both local and dashboard-based alerts are available.

Applications

- Monitoring of grain storage in PDS warehouses (godowns).
- Smart warehouse management for stored rice, wheat and pulses.
- Grain storage by farmers and other storage operators who currently rely on periodic manual checks.

The limitations and boundaries of the system are described in Section 1.2.8.

3.3 SYSTEM ARCHITECTURE

Architectural Overview

The system follows a modular architecture in which each stage passes its output to the next: sensing, processing in the ESP32, data transmission, AI analysis, risk prediction and alerting. Figure 1 shows the overall workflow of the proposed system as presented in the project presentation.

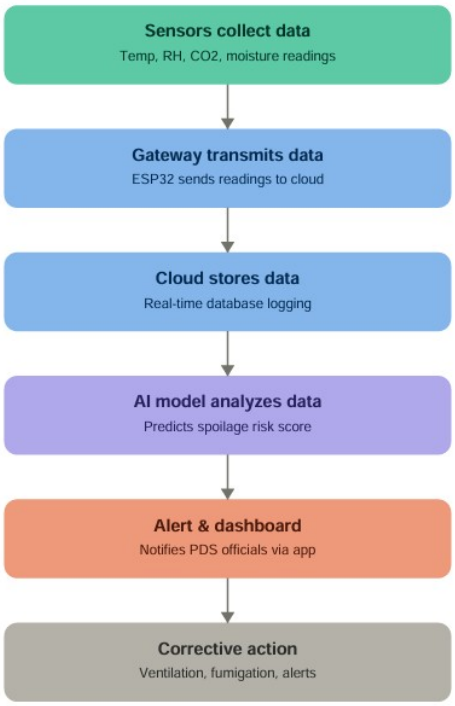


Figure 1 – System architecture and workflow of the proposed system

Architectural Flow

Step 1: Sensing. The DHT22 measures temperature and humidity, the moisture sensor measures grain moisture content, and the MQ135 detects air quality and harmful gases in the storage area.

Step 2: ESP32 processing. The ESP32 microcontroller collects the data from all sensors, performs initial processing and filtering, and prepares the data for transmission.

Step 3: Data transmission. The processed sensor data are transmitted through Wi-Fi to the monitoring dashboard (and, in the proposed architecture, to a cloud server) for real-time access.

Step 4: AI analysis and risk prediction. A Python-based AI/ML model analyzes the incoming sensor data, identifies patterns associated with grain deterioration, and classifies the storage condition as Safe, Moderate Risk or High Risk.

Step 5: Alert and dashboard. According to the predicted risk level, alerts are generated through the LCD display, LED indicators, buzzer alarm and the web dashboard. In the architecture diagram, the final stage is corrective action such as ventilation or fumigation, which is carried out by warehouse staff in response to the alerts; the prototype itself does not perform these actions.

Data Flow

Level 0 (context): Grain storage environment → Smart grain storage monitoring system → Risk status, recommendations and alerts for warehouse staff.

Level 1: Sense → Process (ESP32) → Transmit (Wi-Fi) → Analyze and classify (AI model) → Display and alert (dashboard, LCD, LEDs, buzzer).

3.4 HARDWARE REQUIREMENTS

The prototype uses the hardware components listed in Table 2 and shown in Figure 2.

Table 2 – Hardware components and their functions

S.No.	Component	Function in the project
1	ESP32 development board	Main microcontroller; reads the sensors, processes the data and provides Wi-Fi communication
2	DHT22 sensor	Temperature and humidity monitoring
3	MQ135 sensor	Gas and air-quality monitoring
4	Grain moisture sensor	Grain moisture detection
5	16×2 LCD (I2C)	Displays sensor values and risk status
6	Green, yellow and red LEDs	Low, medium and high risk indication
7	Buzzer	High-risk alert
8	Breadboard and jumper wires	Circuit connections

S.No.	Component	Function in the project
9	Power bank (5 V)	Portable power supply

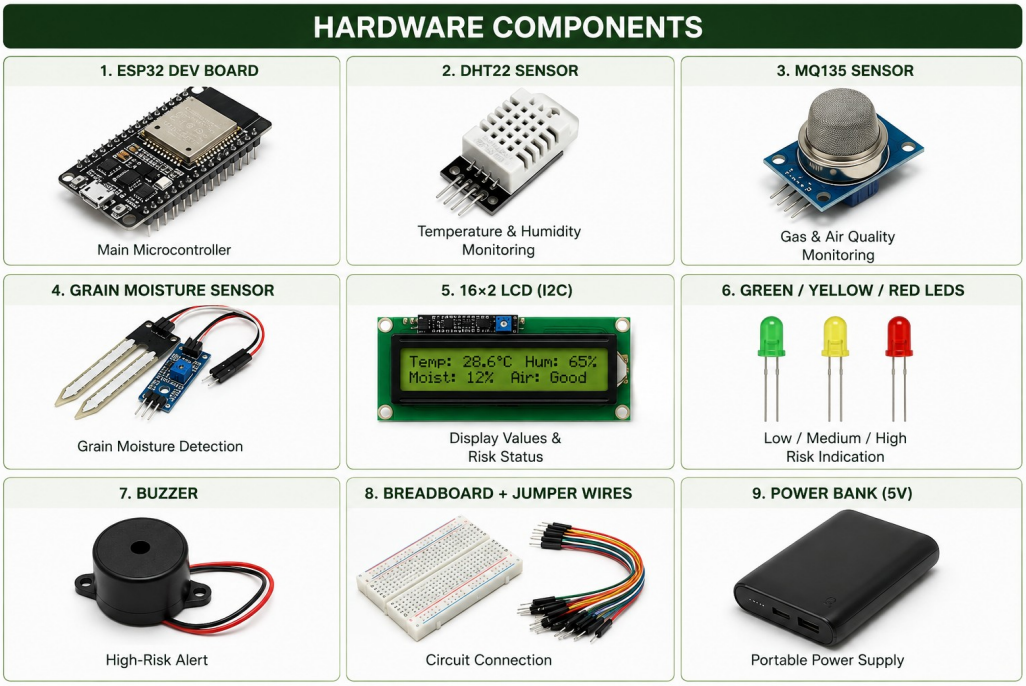


Figure 2 – Hardware components of the proposed system

Hardware Justification

The components are commonly available and inexpensive, which supports the objective of a low-cost and scalable monitoring system. The ESP32 combines processing capability with built-in Wi-Fi, so a separate communication module is not required. Each sensor measures a different indicator of storage condition, and the LCD, LEDs and buzzer give immediate local feedback that does not depend on network access.

3.5 SOFTWARE REQUIREMENTS

The software tools used in the project are listed in Table 3.

Table 3 – Software tools and their usage

S.No.	Software	Description / purpose	Usage in the project
1	Arduino IDE	Integrated development environment used to write, compile and upload code to the ESP32; also used for sensor integration and data reading	ESP32 programming and sensor integration
2	Python	High-level programming language used for developing AI/ML models to predict spoilage risk from sensor data	AI model development for spoilage prediction

S.No.	Software	Description / purpose	Usage in the project
3	Google Colab	Cloud-based Jupyter notebook environment for data preprocessing, AI model training and evaluation	Dataset preprocessing and AI training
4	Web dashboard	Web-based dashboard built with CSS and JavaScript to visualize real-time sensor data, alerts and analytics	Real-time IoT dashboard
5	Microsoft Excel / CSV	Used to store and manage sensor data in tabular format for analysis and model training	Storing the sensor dataset for AI training

Software Modules

- Embedded firmware: reads the DHT22, moisture and MQ135 sensors, drives the LCD, LEDs and buzzer, and handles Wi-Fi communication (Arduino IDE).
- AI module: a Python-based model trained in Google Colab that classifies the storage condition and supports the recommendations.
- Web dashboard: displays the condition status, sensor readings, condition analysis, recommendations, system status and operating mode.
- Dataset module: sensor values and labels stored in tabular (Excel/CSV) form for training.

The proposed system combines low-cost sensing, ESP32-based data collection, and an AI-based risk-classification stage with a dashboard and local alerts. The next chapter describes the methodology and the implementation of each module.

CHAPTER 4
IMPLEMENTATION

4.1 METHODOLOGY

The methodology defines the stages followed to design and implement the system, from monitoring the storage environment to alerting the user. The system follows a seven-stage process in which the output of each stage is the input of the next.

System Development Methodology

Stage 1 – Grain storage monitoring. In PDS warehouses, the storage conditions of rice, wheat and pulses are monitored continuously to maintain grain quality and safety.

Stage 2 – Sensor data collection. The DHT22 measures temperature and humidity, the moisture sensor measures grain moisture content, and the MQ135 detects air quality and harmful gases.

Stage 3 – ESP32 processing. The ESP32 collects the data from all sensors, performs initial processing and filtering, and prepares the data for transmission.

Stage 4 – IoT data transmission. The processed data are transmitted through Wi-Fi to the monitoring dashboard (and, in the proposed architecture, a cloud server) for real-time access.

Stage 5 – AI analysis. A Python-based AI/ML model analyzes the incoming sensor data and identifies patterns associated with grain deterioration and spoilage.

Stage 6 – Spoilage risk prediction. The model classifies the storage condition into three risk levels, as summarized in Table 5.

Stage 7 – Alert system. Based on the predicted risk level, alerts are generated through the LCD display, LED indicator, buzzer alarm and the web dashboard. A mobile application is part of the proposed future enhancements (Section 5.2).

Dataset

The dataset contains temperature, humidity, gas and grain-moisture sensor values collected from grain storage conditions, as listed in Table 4. Each record is labeled as Safe, Warning or Danger based on grain quality. The dataset is stored in tabular form (Excel/CSV), preprocessed in Google Colab, and used to train the AI model to predict the risk of grain spoilage. Live sensor data from the ESP32 are then compared with the trained model for prediction, and the model provides recommendations such as Dry Grains, Improve Ventilation or Inspect Storage.

Table 4 – Attributes of the dataset

S.No.	Attribute	Type
1	Temperature	Input feature (sensor value)
2	Humidity	Input feature (sensor value)
3	Gas (MQ135)	Input feature (sensor value)
4	Grain moisture	Input feature (sensor value)
5	Condition label (Safe / Warning / Danger)	Output class

Risk Categories

Table 5 – Spoilage risk categories and alert indications

Risk level	Meaning	Local indication in the prototype
Safe	Normal storage condition	Green LED (SAFE)
Moderate Risk	Attention required	Yellow LED (WARNING)
High Risk	Immediate action required	Red LED (DANGER) with buzzer alert

In the prototype, the risk levels Safe, Moderate Risk and High Risk correspond to the labels SAFE, WARNING and DANGER used on the indicator panel and in the dataset.

4.2 IMPLEMENTATION

The implementation consists of a hardware module, an ESP32 processing and communication module, an AI risk-prediction module, an alert module and a web dashboard module. Each is described below.

4.2.1 Hardware and Sensor Module

The DHT22, grain moisture sensor and MQ135 gas sensor are connected to the ESP32 development board through a breadboard and jumper wires. The prototype is housed in a model storage cabinet that includes a transparent-door storage chamber, a shelf of model grain packets, an LCD display, three status LEDs labeled SAFE, WARNING and DANGER, and a buzzer labeled ALERT, as shown in Figure 3. A 5 V power bank provides portable power.



Figure 3 – Prototype of the Smart Grain Storage System

4.2.2 ESP32 Processing and Data Transmission Module

The ESP32 is programmed using the Arduino IDE. The firmware reads the sensor values, performs initial processing and filtering, and prepares the data for display and transmission. The ESP32 connects over Wi-Fi so that the readings can be viewed on the monitoring dashboard, which the results of this project present as an ESP32 local monitoring and control dashboard.

4.2.3 AI-Based Spoilage Risk Prediction Module

The AI/ML model is developed in Python and trained in Google Colab using the labeled sensor dataset described in Section 4.1. The model takes the temperature, humidity, gas and grain-moisture values as input and predicts the storage condition as Safe, Moderate Risk or High Risk

(Safe, Warning or Danger in the dataset labels). The prediction is accompanied by a condition analysis and a recommendation such as Dry Grains, Improve Ventilation or Inspect Storage.

4.2.4 Alert and Local Display Module

The 16×2 I2C LCD displays the sensor values and the risk status; the prototype photograph shows the message “Smart Grain System – All Conditions Normal”. The green, yellow and red LEDs indicate low, medium and high risk respectively, and the buzzer provides an audible alert for the high-risk condition.

4.2.5 Web Dashboard Module

The web dashboard is titled “Smart Grain Storage – AI & IoT based grain storage condition monitoring”. Its main elements, as seen in the dashboard screenshots (Figures 4 and 5), are:

- An overall condition banner (for example, NORMAL).
- Sensor cards for temperature, humidity, gas level and moisture, each with a level bar.
- A Condition Analysis panel and a Recommendations panel.
- A Sensor Details panel showing the MQ-135 raw and digital values and the moisture raw value.
- A System Status panel showing the operating mode, the state of the green, yellow and red LEDs, and the buzzer.
- A System Control panel with Auto Mode and Manual Mode selection and LED on/off controls.

CHAPTER 5
CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

The project “Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction for PDS using AI and IoT” addresses the problem of grain spoilage in PDS storage, where periodic manual inspection cannot provide continuous information or early warning. A low-cost prototype was built around an ESP32 microcontroller with a DHT22 temperature and humidity sensor, a grain moisture sensor and an MQ135 gas sensor. Readings are shown on an LCD and on a web dashboard, status is signaled through green, yellow and red LEDs and a buzzer, and an AI/ML model developed in Python and Google Colab classifies the storage condition into risk levels with recommendations.

The results presented in this report demonstrate the working prototype and its dashboard under a normal storage condition. A quantitative evaluation of the prediction model and testing of the Moderate Risk and High Risk conditions are not included in the material on which this report is based; these are the main points to be evaluated before operational use. Within these limits, the

project shows that sensing, wireless transmission, dashboard monitoring and AI-based risk classification can be combined in an affordable system that supports earlier preventive action, reduced manual inspection and better grain-storage management in PDS warehouses.

5.2 FUTURE WORK

The system can be enhanced with additional smart features to improve monitoring and warehouse management:

- Develop a mobile application for real-time monitoring from anywhere.
- Store sensor data in cloud storage for long-term analysis and report generation.
- Add GSM/SMS and notification alerts for farmers and warehouse managers.
- Integrate a camera-based grain inspection system to detect fungal growth and damaged grains.
- Monitor multiple grain storage units using a single dashboard.
- Add data history and graphical reports to analyze storage conditions over time.
- Evaluate and validate the AI model with larger, multi-grain data from real PDS warehouses, as identified in the research gaps in Chapter 2.

5.3 RESULTS

The implemented prototype is shown in Figure 3. With all conditions normal, the LCD of the prototype shows “All Conditions Normal” and the green SAFE indicator is lit. The web dashboard of the prototype is shown in Figures 4 and 5.

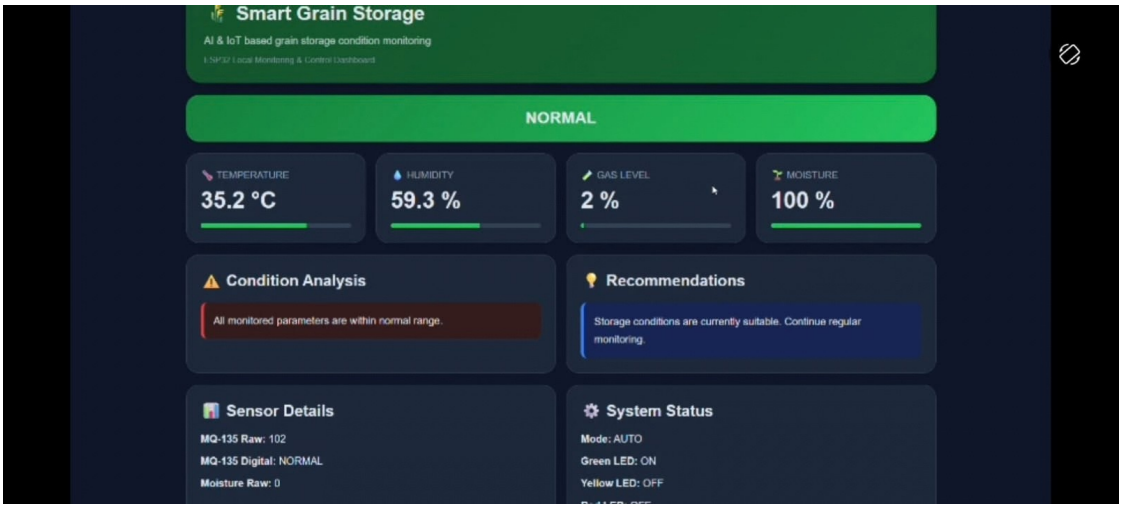


Figure 4 – Web dashboard: condition banner and sensor readings

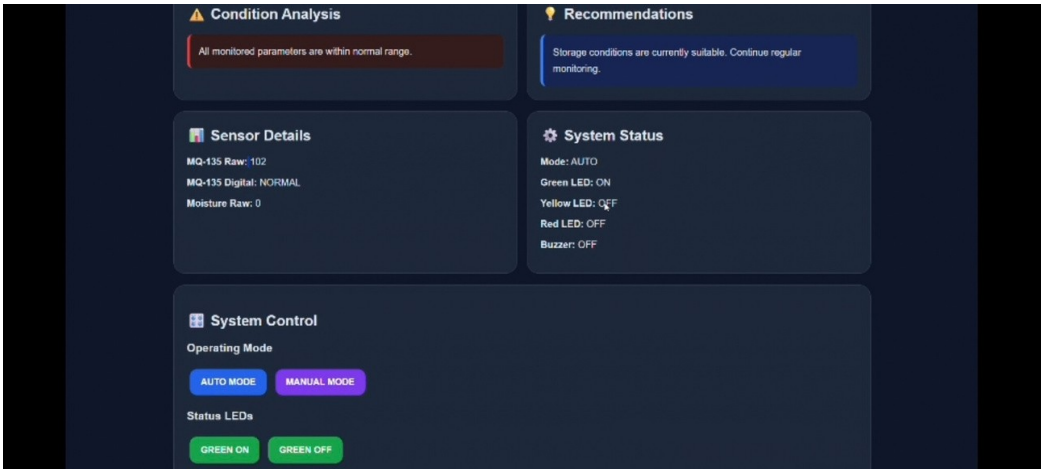


Figure 5 – Web dashboard: condition analysis, system status and system control

Table 6 lists the values displayed on the dashboard in the captured screenshots.

Table 6 – Values displayed on the dashboard in the sample output

Item	Displayed value
Overall condition	NORMAL
Temperature	35.2 °C
Humidity	59.3 %
Gas level	2 %
Moisture	100 %
MQ-135 raw / digital	102 / NORMAL
Moisture raw	0
Condition analysis	All monitored parameters are within normal range.
Recommendation	Storage conditions are currently suitable. Continue regular monitoring.
Mode	AUTO
Green LED / Yellow LED / Red LED / Buzzer	ON / OFF / OFF / OFF

Discussion

The sample output shows that sensor readings reach the dashboard, that the condition is summarized in a status banner with an analysis and a recommendation, and that the LED and buzzer states are reported consistently with the NORMAL condition. This confirms the end-to-end operation of the monitoring and alert path under a normal storage condition. Performance measures of the prediction model, and outputs for the Moderate Risk and High Risk conditions, would complete the evaluation and are not part of the results reported here.

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APPENDIX – CONFERENCE PARTICIPATION CERTIFICATES

This project was presented at the AICTE Sponsored International Conference on Artificial Intelligence-Driven Innovations in Computer, Electrical, Mechanical and Electronics Engineering (ICAI-CEME 2026) at SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES.

SMART GRAIN STORAGE CONDITION MONITORING AND SPOILAGE RISK PREDICTION FOR PUBLIC DISTRIBUTION SYSTEM (PDS) USING AI AND IoT

Gokula Kannan M 1, Umesh R 2, Vignesh S 3 , S.J.kavitha 4

Department of Artificial Intelligence and Machine Learning (AIML), Vels Institute of Science, Technology & Advanced Studies (VISTAS), Chennai, India

Abstract:

Food grain spoilage during storage is a major challenge in the Public Distribution System (PDS) due to environmental factors such as high temperature, humidity, excess moisture, and poor air quality, which reduce grain quality and lead to food wastage and economic loss. This project proposes a Smart Grain Storage Condition Monitoring and Spoilage Risk Prediction System using Artificial Intelligence (AI) and the Internet of Things (IoT) to continuously monitor storage conditions and provide early warnings before spoilage occurs. The proposed system uses an ESP32 microcontroller integrated with a DHT22 temperature and humidity sensor, grain moisture sensor, and MQ135 gas sensor to collect real-time environmental data from grain storage facilities. The collected data is transmitted to an IoT cloud platform for continuous monitoring and analysis. A Machine Learning model processes the sensor data and predicts the spoilage risk by classifying the storage condition as Safe, Moderate Risk, or High Risk, enabling warehouse authorities to take timely preventive actions. The proposed solution aims to reduce food wastage, improve grain storage efficiency, minimize manual inspection, and support smart warehouse management in PDS godowns through a low-cost, scalable, and intelligent monitoring system.

Keywords: Artificial Intelligence, Internet of Things, Smart Grain Storage, Spoilage Risk Prediction, ESP32, Grain Moisture Monitoring, Public Distribution System

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
ISBN No.: 978-93-344-7370-4



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VELS UNIVERSITY (VISTAS)

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SMART GRAIN STORAGE CONDITION MONITORING AND SPOILAGE RISK PREDICTION FOR PUBLIC DISTRIBUTION SYSTEM (PDS)

held during 21.10.2026 & 22.10.2026 organized by the Institute of CSE, SIMATS Engineering, SIMATS, Thandalam, Chennai. Your presence and engagement have contributed significantly to the discussions and knowledge-sharing on AI-driven Engineering Innovations.

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ICAI-CME 2026

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